ANALYSTLAB AFRICA

Week 6 Statistical Analysis Report

HealthConnect Rwanda - Data Science Track

Takudzwa Jeffrey Gumbo (TJ) | Data Science Track

Synthetic educational healthcare data. Analytical outputs are not clinical advice or diagnosis.

1. Methods

Significance level alpha = 0.05. Non-parametric tests are used where rating/billing distributions are not assumed normal, and a chi-square test is used for categorical admission status by age group.

2. Hypothesis tests

Test	Method	Statistic	p-value	Decision
H1	Kruskal-Wallis	13.800	0.7952	Fail to reject H0
H2	Chi-square independence	10.653	0.0308	Reject H0
H3	Mann-Whitney U	2332992.500	0.5438	Fail to reject H0

3. Interpretation

H1 - Satisfaction across facilities: p=0.7952. There is no statistically significant evidence of different satisfaction distributions across FacilityIDs in the matched detailed sample.

H2 - Admission by age group: p=0.0308. The null of independence is rejected at 5%, indicating that admission occurrence varies across the defined age groups in this synthetic detailed sample. This is an association, not a clinical causal conclusion.

H3 - Billing by recorded chronic-condition status: p=0.5438. There is no statistically significant evidence that billing distributions differ between records with and without a recorded chronic condition in the matched sample.

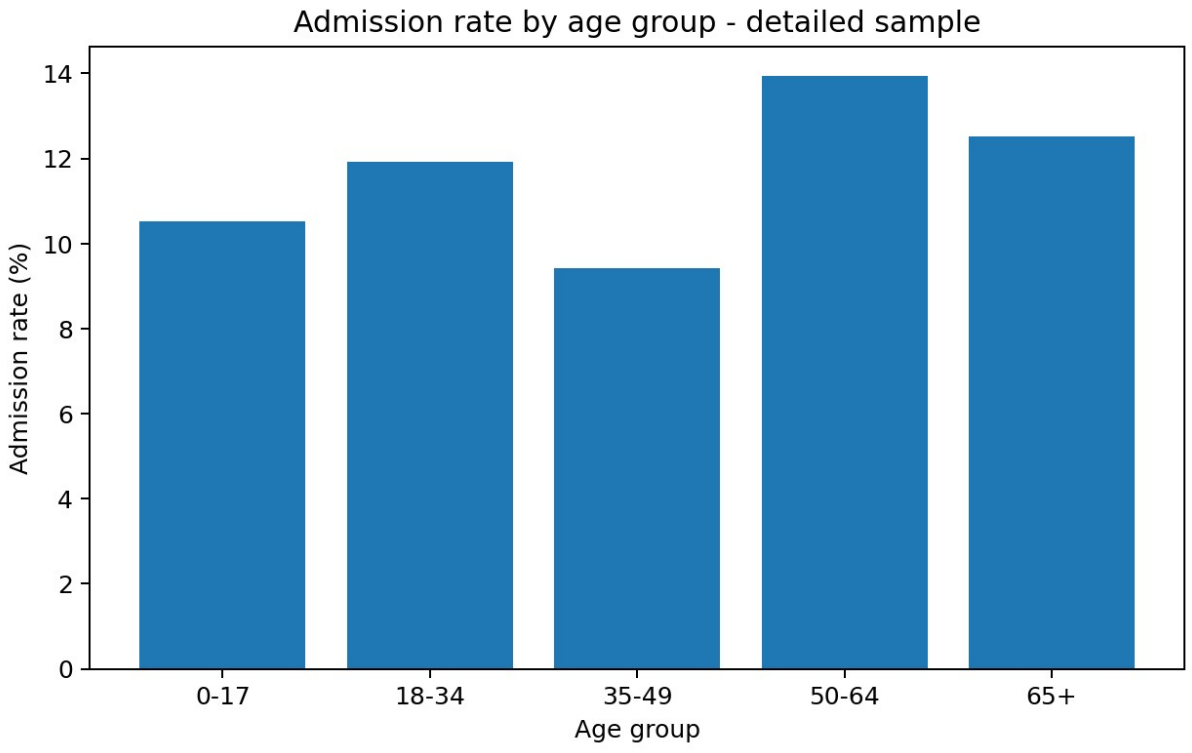


Figure 1. Admission rates by age group in the detailed sample.

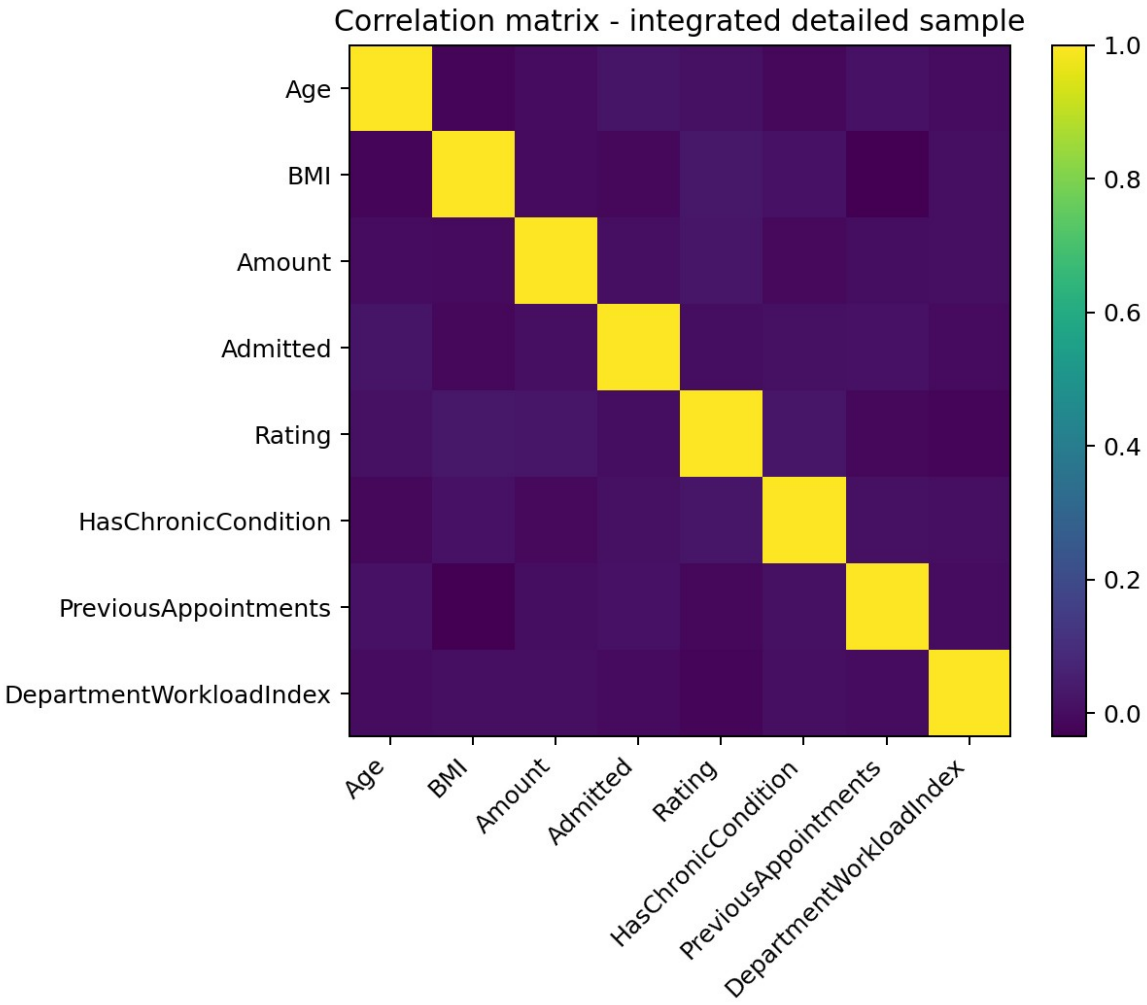


Figure 2. Correlation matrix for selected analytical variables.